

Stay-Deepfake: A Case for Ignoring Background Features to Prevent Shortcut Learning

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Abstract—Image manipulation techniques are more realistic today using deep neural network techniques than they were early 2020. This have led to many sectors worrying about spreading of misinformation. This project compares two state-of-the-art deep learning pipelines to an improved pipeline of a diffusion based detector. We retrain the last layers of a baseline network such that the relationships only link to the parts of the image that are manipulated. The improved pipeline were able to increase in accuracy and force the model to focus on the most important feature, the face.

Index Terms—DeepFakes, Face Manipulation, Fake Detection, Convolutional Neural Networks, Deep Learning

I. INTRODUCTION

The rapid evolution of deep generative models has made the synthesis of highly realistic facial manipulations a pressing threat to multimedia integrity. Consequently, the development of robust deepfake detection frameworks has become a critical priority. Current state-of-the-art detection models generally rely on deep convolutional neural networks (CNNs) [1] or diffusion-based architectures [2] to extract micro-texture discrepancies and blending artifacts. However, evaluating these models across real-world, unconstrained environmental distributions reveals a systemic vulnerability.

This research initially set out to benchmark two distinct state-of-the-art paradigms against our own proposed framework. The first baseline relies on a standard CNN-based architecture, while the second evaluates a diffusion-based detection approach. To rigorously assess their decision-making processes, we integrated explainable AI techniques [3], specifically Grad-CAM[4] and LIME[5]. Visualizing the attention maps of these models exposed a glaring reliance on shortcut learning, also known as lazy learning [6].

Further inspection revealed that the CNN-based model[1] achieved artificially high classification scores through severe data leakage. For manipulated images, the network correctly focused on the facial region; however, for authentic images, its attention erroneously shifted to the background context. Because the original authors did not employ explainability metrics, this critical limitation went unnoticed. Conversely, the diffusion-based model, which was originally trained on tightly cropped data, failed significantly when tasked with generalizing to datasets containing diverse backgrounds, such as FaceForensics++ (FF++)[7]. This performance drop highlights

a foundational crisis in fake media classification: detectors are overwhelmed by real-world environmental signals, causing them to classify images based on background distributions rather than actual forgery artifacts [8].

Therefore, the primary challenge addressed in this work is mitigating data leakage and preventing networks from associating real images with background features and fake images with facial features. To resolve this, our methodology draws inspiration from the “Stay-Positive” paradigm [9]. Although originally designed for detecting fully synthesized fake images rather than localized deepfakes, the core mathematical philosophy directly addresses our identified limitation. The Stay-Positive approach operates on the premise of isolating generative traces by forcing the network to ignore real-world features.

We applied this theoretical foundation to design our proposed detection framework, Stay-Deepfake, which is built upon the DiffConvNet architecture⁴. The standard DiffConvNet model suffers heavily from lazy learning and fails completely on background-heavy datasets due to overwhelming environmental noise¹⁰. To counteract this, Stay-Deepfake adopts the Stay-Positive methodology by freezing the feature extraction layers and transforming the final classification layer into a Rectified Linear Unit (ReLU) layer [9]. This specific modification effectively zeros out all negative values associated with real features while preserving the positive values associated with generative manipulation.

By passing our dataset through a standardized facial cropping algorithm and subsequently processing it through Stay-Deepfake, we successfully trained the model to mathematically ignore environmental contexts. This forces the network to focus exclusively on the facial region of interest, yielding a highly robust and generalized deepfake detection framework.

A. Workload Distribution

The successful execution of this project was the result of a highly collaborative and equal effort between both authors. Rather than dividing the project into isolated tasks, all phases of the research were conducted jointly. Both Jeet Purohit and Alexander Alfaro contributed equally to the conceptualization of the Stay-Deepfake architecture, the implementation and testing of the baseline CNN and diffusion models, the data

preprocessing pipelines, the extraction of explainability metrics via Grad-CAM and LIME, and the drafting and reviewing of this final report.

II. PROBLEM DEFINITION

The fundamental challenge in modern deepfake detection is not the inability of neural networks to find patterns, but rather their tendency to find the wrong ones. This phenomenon is formally known as shortcut learning or lazy learning [6]. When tasked with distinguishing between authentic media and deepfakes, deep learning models naturally default to minimizing their loss function through the easiest available environmental cues. Instead of learning the complex, microscopic blending artifacts that actually indicate a manipulated face, these networks frequently memorize background contexts, global camera compression rates, or specific chin boundary lines present in the training datasets[8].

This shortcut learning issue is uniquely severe in the deepfake domain compared to general fake image detection. In a fully synthesized image, the entire composition is artificially generated. However, a deepfake is inherently localized; a manipulated face is seamlessly inserted into a completely authentic, real-world background. This coexistence of real and fake elements within the exact same frame introduces critical data leakage. As observed during our baseline testing with explainability tools, models easily fall into the trap of associating the authentic class with specific background distributions and the fake class with facial regions. Consequently, they completely fail to generalize when applied to real-world scenarios containing unseen backgrounds.

Therefore, the specific problem we set out to solve is determining whether a mathematical framework designed for fake image detection can be successfully adapted to overcome the localized challenges of deepfake detection. The Stay-Positive theory recently demonstrated that neural networks can be forced to ignore real-world features when classifying fully synthetic images [9]. Our primary objective in this research is to investigate if applying this theory yields the same corrective effect on localized deepfakes. Specifically, we define our problem as testing whether we can construct a framework, Stay-Deepfake, that mathematically blinds the network to the authentic background, thereby forcing the model to classify an image as fake if and only if it detects generative artifacts within the facial region.

III. RELATED WORK

This section explains different face manipulation techniques used in the field. Most of the information and references in this section are extracted through the article: Deepfakes and Beyond [10].

- **Entire face synthesis:** Generative models can create completely realistic, non-existent human faces. This is usually created through powerful GAN models, e.g.,

StyleGAN approach[11]. The article Deepfakes and Beyond explains that this type of manipulation is of great use in sectors such as 3D-modelling or in video games, but it has some harmful applications such as creating realistic fake profiles on different social media platforms to spread misinformation.

- **Identity swap:** If created using deep networks called Deepfakes[10]. This involves replacing the face of a target individual with the face of another source person. Image frames are provided to an autoencoder, then a decoder of some other image is used to generate the fake image[12]. This type of manipulation could possibly benefit in face enhancing, but it could be used in creating pornographic videos, hoaxes and financial fraud etc. This is a problem that encouraged the creation of this project.
- **Attribute Manipulation:** Modifying specific facial attributes such as hair color, age, gender, or adding glasses[13]. Examples of use for this technique are: consumers could try different products such as cosmetics, makeup or clothes in a virtual environment[10] but also potentially spread false information of high ranking profiles.
- **Expression swap:** Transferring facial expressions from a source video to a target face while keeping the target's identity. Some swaps are made through the known GAN model architectures, STGAN[14] or Face2Face[15]. applying this technique could have some serious consequences e.g., Mark Zuckerberg saying things he never said, shown in a Youtube video ¹.
- **Manipulation detection:** There are machine learning frameworks capable of spotting manipulation techniques through artifact analysis. One method is proposed in the article Stay-Positive[9] which we have put to use in our proposed method. We will give a brief overview of the proposed detection methods for the above manipulation techniques. For the entire face synthesis, some researches hypothesized that the difference in colour between a real image and a fully synthetic image are notable. They proposed a detection model based on colour features[16]. There are also detection methods in the area of identity swap techniques. Celeb-DF[17] suggest a state of the art deepfake dataset for benchmarking detection systems and explains the challenges in the area. Attribute manipulation techniques could have subtle features but its still possible to detect. One such research is proposed by Fake spotter[18] which analyse the behaviour of neurons between real and fake images. MesoNet[19] is an approach where they tested against Face2Face[15] and got great results according to the article deepfakes and beyond [10].

IV. PROPOSED METHOD

We trained two architectures CNN[1] and diffusion based model[2] and saw that the models prediction was very much based on the background. We can see the Grad-cam explanation on a fake image1. We saw similar explanation using the CNN model[1].

¹<https://www.bbc.com/news/technology-48607673>



Fig. 1. Grad-Cam shows where the diffusion based model[2] has based its prediction on.

We tried retraining on cropped images which will be mentioned in detail in experimental methods. We need another method for forcing the model to focus more on the fake features. we have in earlier courses analysed and presented a method that was proposed in the article Stay-Positive[9]. We hypothesized that if we could make the diffusion model ignore the real features i.e. stop predicting based on how much fake it is compared to how real it is, maybe the model could focus more on the faces than on the background which is both real in the manipulated and real dataset images. Our work builds on retraining the relationships in the last layer, making the model only focus on the fake features and make its prediction.

V. EXPERIMENTAL SETUP

To validate the effectiveness of the Stay-Deepfake architecture against shortcut learning, we established a rigorous experimental framework. This section outlines the hardware, software technologies, and preprocessing configurations utilized to ensure reproducible results.

A. Implementation and Hardware Details

All models, data extraction pipelines, and preprocessing scripts were implemented using Python 3.1x. The neural network architectures, including the CNN baseline, DiffConvNet, and the proposed Stay-Deepfake model, were constructed and trained utilizing the PyTorch deep learning framework.

To handle the computationally intensive tasks of diffusion modeling and batch processing, all training and evaluation phases were accelerated using an NVIDIA GeForce RTX 5080 GPU equipped with 16 GB of VRAM, running on CUDA 13.2. It should be noted that reproducing these experiments on hardware with lower memory capacities may require a proportional reduction in batch size to prevent out-of-memory errors. For computer vision preprocessing tasks, we relied on OpenCV and the Python Imaging Library (PIL). Furthermore, to evaluate the models for lazy learning, we integrated explainable AI modules, specifically Grad-CAM and LIME, into our PyTorch testing environment.

B. Datasets

For our experiments, we selected the FaceForensics++ (FF++) dataset. We chose this dataset because of its

well-known benchmark status and its diverse mix of manipulation technologies. The dataset fundamentally consists of 1,000 authentic videos (original) and 5,000 manipulated counterparts spread across five distinct fake generation categories: Deepfakes, Face2Face, FaceShifter, FaceSwap, and NeuralTextures.

To strictly prevent data leakage and avoid identity overlap, we avoided random frame-level splitting. Instead, we performed a completely isolated video-level split. Exactly 700 distinct videos from the original folder and 700 corresponding videos from the fake categories were explicitly assigned to the training set. The remaining 300 unique videos were strictly reserved for the validation set.

From these partitioned videos, frames were sampled at a 2-second interval for a maximum duration of 20 seconds, yielding up to 10 pictures max per video. After processing, the final training dataset consisted of 14,690 images (5,843 original, 8,847 fake). The strictly isolated validation dataset consisted of 9,491 images (2,522 original, 6,969 fake).

C. Dataset Processing

To ensure the models focused strictly on the appropriate regions of interest, we implemented an automated facial extraction and validation pipeline.

First, frames were extracted from our designated training and validation videos. To isolate the faces, we utilized OpenCV's Haar Cascade classifier. The algorithm identified the largest bounding box corresponding to a face and applied a 20% spatial margin around it before cropping. This margin was intentionally included to capture peripheral background details for our baseline tests. All cropped regions were subsequently resized to a standard resolution of 224×224 pixels.

However, automated cropping algorithms can occasionally misidentify objects or produce false positives, especially in heavily manipulated videos. To prevent degraded data from compromising our training sets, we engineered an automated cleanup pass using OpenCV. This validation script analyzed every outputted crop by running a secondary Haar Cascade detection. The script mathematically calculated the area of the detected face relative to the total area of the cropped image. If the classifier failed to detect a face, or if the detected facial area constituted less than 8% of the total image area, the crop was flagged as a false positive and automatically removed from the dataset. This rigorous two-step pipeline ensured our final dataset consisted exclusively of high-quality, accurately localized facial images.



Fig. 2. Example of the final image after going through processing

D. CNN-Based Model Baseline

To establish a state-of-the-art performance baseline and demonstrate how traditional spatial networks remain susceptible to shortcut learning, we implemented the lightweight Convolutional Neural Network (CNN) architecture proposed by Kapoor and Singh for deepfake detection [1].

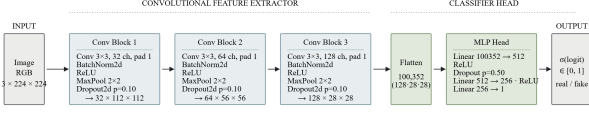


Fig. 3. Architecture of the Kapoor and Singh CNN baseline model

Following the methodology outlined in their study, the network consists of a three-block feature extractor designed for efficiency. Each extraction block applies a two-dimensional convolution, batch normalization, a Rectified Linear Unit (ReLU) activation to capture non-linear relationships, and max-pooling to achieve spatial dimensionality reduction. To mitigate overfitting, a 10% spatial dropout rate was applied after every convolutional block, aligning with the authors’ core architectural design. The extracted features were then flattened and passed through two fully connected layers containing 512 and 256 neurons, regulated by a 50% dropout constraint, before reaching the final single-neuron binary classification output.

Training Configuration and Deviations: While we preserved the core CNN architecture proposed by Kapoor and Singh, we intentionally deviated from their original training hyperparameters. This adjustment was necessary to optimize the network’s convergence behavior for the specific complexities of the FaceForensics++ (FF++) dataset and to accommodate our hardware environment. The exact training configuration utilized in our baseline experiments is detailed in Table I.

Parameter	Configuration Value
Maximum Epochs	50
Batch Size	128
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)
Base Learning Rate	0.0001
Loss Function	BCEWithLogitsLoss (dynamic pos_weight)
Early Stopping Patience	15 Epochs
Checkpoint Interval	10 Epochs

TABLE I
CUSTOM TRAINING HYPERPARAMETERS FOR THE CNN BASELINE ON THE FF++ DATASET.

To account for class imbalances within the extracted dataset, the Binary Cross Entropy loss was dynamically scaled using a positive weight multiplier. To ensure the model continually refined its weights as it approached convergence, we implemented a learning rate scheduler (`ReduceLROnPlateau`) that dynamically reduced the learning rate by a factor of 0.3 if the validation loss plateaued for 3 consecutive epochs. Finally, the training data was subjected to an augmentation pipeline

consisting of random horizontal flips (50% probability), 15-degree spatial rotations, and 20% brightness jittering prior to standard 224×224 resolution resizing.

E. DiffConvNet Architecture

To evaluate the capability of diffusion-driven feature extraction, we implemented the DiffConvNet framework proposed by Pintelas and Livieris [2]. Unlike traditional classifiers, this network integrates a diffusion process directly within a Convolutional Neural Network (CNN) backbone to isolate complex generative artifacts.

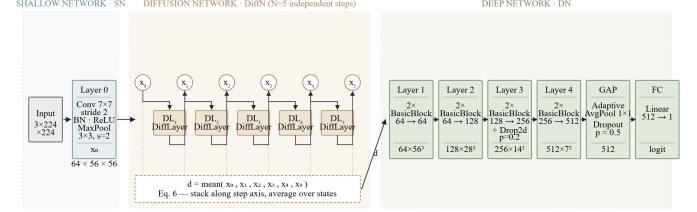


Fig. 4. Architecture of DiffConvNet

As detailed in the original study [2], the architecture utilizes a pre-trained ResNet-18 backbone that is conceptually partitioned into two segments: a Shallow Network (SN) and a Deep Network (DN). A specialized Diffusion Network (DiffN) is systematically inserted between them. In this sequential pipeline, the SN extracts initial spatial features, the DiffN iteratively diffuses these encoded representations to expose microscopic inconsistencies, and the DN processes the averaged diffusion states to generate the final classification logits.

Architectural Limitations and Data Formatting: While the original authors demonstrated the efficacy of DiffConvNet, they did not evaluate their architecture against full-frame images containing complex, real-world backgrounds. During our experiments, we discovered that the model inherently cannot process full-image environments without failing. Because the original study omitted testing under these realistic conditions, the authors failed to identify the network’s severe vulnerability to background interference. Consequently, to allow the model to function at all, it was mandatory to utilize our extracted dataset rather than raw frames. All inputs were restricted to the facial bounding box with a 20% spatial margin and standardized to a resolution of 224×224 pixels. However, as our explainability tools later revealed, even with this necessary localized cropping, the network still managed to exploit the residual background present in the 20% margins to bypass learning true generative artifacts.

Training Configuration and Deviations: Because DiffConvNet relies on complex internal diffusion states, training must be conducted in two distinct computational phases. We deviated slightly from the original paper’s generalized training loop to optimize the network specifically for the FaceForensics++ (FF++) dataset. The exact training configuration utilized in our experiments is detailed in Table II.

Parameter	Configuration Value
Batch Size	128
Optimizer	Adam (Weight Decay = 1×10^{-4})
Loss Function	Smooth BCE Loss (Label Smoothing = 0.1)
Scheduler	Cosine Annealing
Phase 1: Base Network	Values
Phase 1 Epochs	10
Phase 1 Learning Rate	1×10^{-4}
Phase 2: Diffusion Network	Values
Phase 2 Epochs	50
Diffusion Layers (DiffN) LR	5×10^{-5}
Deep Network (Layer 4) LR	1×10^{-5}
Classification Head (FC) LR	1×10^{-4}

TABLE II
CUSTOM TWO-PHASE TRAINING HYPERPARAMETERS FOR THE
DIFFCONVNET MODEL.

Phase 1 involved fine-tuning the base ResNet backbone to establish a foundation for facial feature extraction. Phase 2 introduced the DiffN layer and trained the complete SN-DiffN-DN pipeline. To maintain gradient stability across the network’s varying depths during this second phase, we applied differential learning rates tailored to each specific block. To account for dataset class imbalances, the Binary Cross Entropy loss was dynamically scaled using a positive weight multiplier, and a 10% label smoothing constraint was applied to prevent overconfident network predictions.

Finally, to force the model to generalize, the training data was subjected to a robust augmentation pipeline consisting of random horizontal flips, 20% random perspective distortion, intense color jittering, and random erasing (20% probability) to simulate occlusions.

F. Stay-Deepfake Architecture

To address the shortcut learning vulnerabilities identified in both the baseline CNN and the standard DiffConvNet, we propose the Stay-Deepfake architecture. This model builds upon the proven diffusion-based feature extraction of DiffConvNet but integrates a novel mathematical constraint during final classification based on the Stay-Positive methodology introduced by Sundara Rajan and Lee [9].

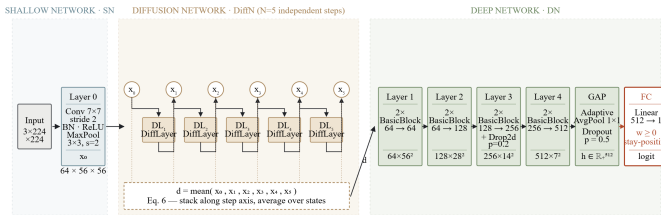


Fig. 5. The Stay-Deepfake architecture

The theoretical foundation of the Stay-Positive constraint argues that a detector should classify an image as a deepfake if and only if it explicitly contains artificial generative

artifacts [9]. Traditional models often assign negative weights to authentic image features (such as standard compression artifacts or unaltered background textures). As a result, the mere presence of these authentic features can artificially lower the deepfake prediction score, causing the model to prioritize background context over the actual manipulation [9].

Architectural Constraints and Weight Clamping: To physically force the network to ignore authentic image features, we fundamentally altered how the final classification head operates. First, because the Deep Network (DN) segment utilizes a pre-trained ResNet backbone, its final residual block naturally concludes with a Rectified Linear Unit (ReLU) activation. This architectural detail guarantees that all feature embeddings passed from the diffusion pipeline to the classifier are strictly non-negative.

To complete the constraint pipeline, we modified the optimizer logic for the final fully connected classification layer. After every backward propagation pass during training, a programmatic clamping function (`clamp_(min=0.0)`) explicitly forces any negative weights to zero. Because both the incoming features and the network weights are strictly non-negative, the model is mathematically prohibited from lowering its prediction score based on authentic contextual data. It can only classify an image as a deepfake by positively identifying manipulation artifacts.

Training Configuration: The Stay-Deepfake model required a specialized three-phase training pipeline. Phases 1 and 2 were identical to the standard DiffConvNet configuration to establish robust spatial and diffusional feature extraction.

In Phase 3, all convolutional and diffusion layers were explicitly frozen to prevent feature drift. The network then extracted the non-negative embeddings for the entire localized dataset, and the isolated classification head was trained under the strict non-negativity constraint. The comprehensive training parameters for this final phase are detailed in Table III.

Parameter	Configuration Value
Phases 1 & 2: Feature Extraction	Values
Architecture	SN-DiffN-DN Pipeline
Phase 1 & Phase 2 Parameters	Identical to DiffConvNet
Phase 3: Stay-Positive Classification	Values
Phase 3 Epochs	15
Phase 3 Batch Size	1024
Phase 3 Optimizer	Adam
Phase 3 Learning Rate	1×10^{-3}
Loss Function	BCEWithLogitsLoss

TABLE III
THREE-PHASE TRAINING HYPERPARAMETERS FOR THE PROPOSED
STAY-DEEPAKE MODEL.

VI. RESULTS

The results are based on the base dataset discussed in experimental setup section[7]. We mostly rely on GRAD CAM [4] and LIME[5] for an explanation, both are explained though Interpretable Machine Learning by Molnar[3]. Applying our method increases accuracy 76% to 77%, AUC 0.84% to 0.85%, recall 77% to 98% and precision decreased from 90%

to 77% during the training phase (figure 6) where we used cropped data. On the validation using full images (figure 7) we see an increase in accuracy; 49% to 73%, similar values in precision 74% and an increase in recall; 47% to 99%. An interesting finding is that our AUC metric decreased on all stages when validating using the full image data. We believe this is a sign that our model has not truly learned to only check for faces but our other metrics such as GRAD-CAM and LIME which are metrics specific for images make us question if AUC is a correct metric for this specific task. AUC describes how well our model guesses right, because we have split the train/test data as 70/30 respectively, we hypothesize that the model only guesses what the deep fakes are on the validation.

	Model	Accuracy (%)	Precision	auc	Recall	F1
0	Base Network	83.94%	0.9229	0.9076	0.8525	0.8863
1	DiffConvNet (Stage 1)	76.22%	0.8952	0.8415	0.7658	0.8255
2	Stay-Positive (Final)	77.34%	0.7720	0.8451	0.9811	0.8641

Fig. 6. Metrics on training images that are cropped forcing the model to learn faces.

	Model	Accuracy (%)	AUC	Precision	Recall	F1
0	Base Network	42.53%	0.5862	0.7971	0.2929	0.4284
1	DiffConvNet (Stage 1)	48.60%	0.5206	0.7383	0.4660	0.5714
2	Stay-Positive (Final)	73.14%	0.5282	0.7350	0.9925	0.8445

Fig. 7. Metrics on validation images that are not cropped. Note that the AUC metric is not present here.

It is much more obvious that our model, after retraining its relationships using Stay-positive[9], are now using the faces to make its prediction.

In figure 8 and 8 we can see the improvement of GRAD-CAM and also how LIME shows the part of the face that is fake. The probability of the prediction is not comfortably high, we can still say that we are on the way.



Fig. 8. Diffusion based approach[2] for a FAKE image. Original image, sum of the five diffusion states, GRAD-CAM, LIME explanation and the prediction shown side by side to show the pipeline of the model



Fig. 9. Retrained using Stay-Positive[9] for a FAKE image. Original image, sum of the five diffusion states, GRAD-CAM, LIME explanation and the prediction shown side by side to show the pipeline of the model

Below two images are from the Real dataset. The interesting part is the LIME explanations. The GRAD-CAM is still centred in on the face but if we look at the LIME explanations its not really checking for the correct places, and it gets worse on real images. This could mean that our method got better at predicting fake images which was our goal. similar to the medical sector, its better that some real images are classed as fake, then fake images classed as real.



Fig. 10. Diffusion based approach[2] for a REAL image. Original image, sum of the five diffusion states, GRAD-CAM, LIME explanation and the prediction shown side by side to show the pipeline of the model

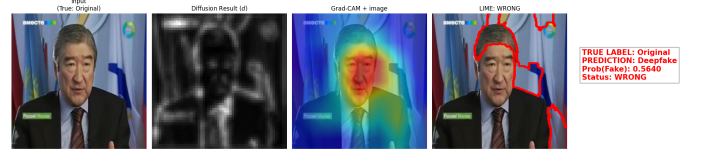


Fig. 11. Retrained using Stay-Positive[9] for a REAL image. Original image, sum of the five diffusion states, GRAD-CAM, LIME explanation and the prediction shown side by side to show the pipeline of the model

VII. DISCUSSION

We put much time on the data preparation stage, all the models changed their metrics and behaviour depending on the dataset and image preprocessing. The Diffusion based model[2] had great performance on a dataset that is very different to FaceForensics++[7]. On many occasions we tried to make the state of the art methods better or change them but we realized that we would have needed the same dataset that each article used for us to get the metrics they got. The details of the data preparation steps for their models could not be found and we had to settle on what the vast majority of researcher do in the data preparation step. If it was easier to exactly recreate their pipeline we could have put much more time in improving our own method and create ablation study.

A. Future work

Addressing the escalating threat of manipulated media requires the continuous enhancement of existing state-of-the-art

models. We believe that integrating novel techniques, specifically the Stay-Positive methodology, is key to developing next-generation deep learning pipelines. By exploring this approach, current architectures can be made significantly more robust against shortcut learning. Consequently, further research into combining these methodologies is essential to tackle this growing problem.

VIII. CONCLUSION

In this research, we leveraged prior knowledge from the fake image detection domain and applied it to a state-of-the-art deepfake detection model to enhance its robustness against spurious features, such as background environments. Furthermore, through the integration of explainable AI techniques, specifically Grad-CAM and LIME, we demonstrated that this methodology successfully forces the model to focus on the relevant facial features.

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